The mean value theorem Let $y = f(x)$ be continuous for $a \le x \le b$ and possess a derivative at each x for $a < x < b$. Then there is at least one number θ between a and b (see Fig. 3.6.1) such that

$$f(b) - f(a) = (b - a)f'(\theta) \qquad a < \theta < b \tag{3.6.1}$$

This is equivalent to stating that at the position θ the slope of the derivative is parallel to the chord joining the two endpoints.

Equation (3.6.1) can sometimes be used to avoid roundoff (and sometimes a lot of machine time). For example,

$$\ln(N + a) - \ln N = a\frac{1}{\theta}$$

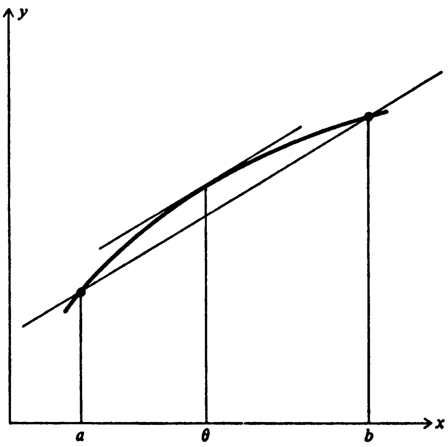


FIGURE 3.6.1

and it is reasonable to pick θ as the midvalue

$$\ln(N + a) - \ln N \approx \frac{2a}{2N + a}$$

Let $N = 100$ and $a = 1$; then

$$\ln 101 - \ln 100 = 4.61512 - 4.60517 = 0.00995$$

Again,

$$\frac{2}{200 + 1} = \frac{1}{100.5} = 0.00995$$

and no logs were evaluated in the computation.

PROBLEMS

Apply the mean value theorem and check the result if numbers are given.

3.6.1 $\sqrt{82} - \sqrt{81}$

3.6.2 $\sin(x + \varepsilon) - \sin x$

3.6.3 $\arctan(N + 1) - \arctan N \qquad N = 10$

3.6.4 $\int_N^{N+1} \ln x \, dx \qquad N = 10$

3.7 SYNTHETIC DIVISION

Polynomials play a central role in computing. Not only do they occur naturally, but everything that the four arithmetic operations (addition, subtraction, multiplication, and division) can produce is the equivalent of one polynomial divided by another. It is important, therefore, to examine with some care the problem of evaluating a polynomial for a given value of the argument x.

The polynomial

$$P(x) = a_N x^N + a_{N-1}x^{N-1} + a_{N-2}x^{N-2} + \cdots + a_1 x + a_0$$

can be written in the "chain" (nested) form

$$\{\cdots[(a_N x + a_{N-1})x + a_{N-2}]x + \cdots a_1\}x + a_0$$

which involves N additions and N multiplications.

The chain method is the same that occurs in the synthetic division process of dividing a polynomial by a linear factor $x - a$. For example, the particular polynomial

$$P(x) = x^4 + 6x^2 - 7x + 8$$

divided by $x - 2$ leads to

$$\begin{array}{rl}
 & x^3 + 2x^2 + 10x + 13 = \text{quotient} \\
\text{Divisor} = x - 2 & \overline{\big)\, x^4 + 0x^3 + 6x^2 - 7x + 8} \\
 & \underline{x^4 - 2x^3} \\
 & \quad 2x^3 + 6x^2 \\
 & \quad \underline{2x^3 - 4x^2} \\
 & \qquad 10x^2 - 7x \\
 & \qquad \underline{10x^2 - 20x} \\
 & \qquad\qquad 13x + 8 \\
 & \qquad\qquad \underline{13x - 26} \\
 & \qquad\qquad\quad + 34 = \text{remainder}
\end{array}$$

which can be written as

$$\frac{P(x)}{x-2} = \frac{x^4 + 6x^2 - 7x + 8}{x-2} = x^3 + 2x^2 + 10x + 13 + \frac{34}{x-2}$$

or as

$$P(x) = \underbrace{x^4 + 6x^2 - 7x + 8}_{\text{polynomial}} = \underset{\substack{\uparrow \\ \text{divisor}}}{(x-2)}\underbrace{(x^3 + 2x^2 + 10x + 13)}_{\text{quotient}} + \underset{\substack{\uparrow \\ \text{remainder}}}{34}$$

When $x = 2$, we get

$$P(2) = 34 = \text{remainder}$$

This is a special case of the remainder theorem.

The remainder theorem If a polynomial $P(x)$ is divided by $x - a$, we get

$$P(x) = (x-a)Q(x) + R$$

where $Q(x)$ is the quotient and R is the remainder, and we have

$$P(a) = R$$

If $R = 0$, then a is a zero of $P(x)$.

The division process can be simplified by noting that the powers of x are place holders and need not be written *provided* zero coefficients are supplied for the missing terms. Furthermore, the quotient need not be written on the top line since it is given by the numbers at the bottom of each column. Finally, we need write not $x - a$, but a, and then add, instead of subtract, in the body of the form. As a result of all these changes we get

$$\begin{array}{r|rrrrl} \text{divisor} \rightarrow 2 & 1 & 0 & 6 & -7 & 8 \\ & & 2 & 4 & 20 & 26 \\ \hline & \multicolumn{4}{r}{\underbrace{1 \quad 2 \quad 10 \quad 13}_{\text{quotient}}} & \boxed{34} = \text{remainder} \end{array}$$

In this form it is clearly the chain method of evaluating a polynomial $P(a)$.

If the quotient is again divided by the same linear factor, we will get another quotient and another remainder r_1. Dividing this new quotient again by the same factor, etc., we find that we are performing exactly the same process that was used to represent a number as a sum of powers of a number base (Sec. 2.7 where we used the base 2), and correspondingly we are representing a polynomial as a sum of powers of the linear factor we were dividing by. Thus we will get

$$P(x) = r_0 + r_1(x-a) + r_2(x-a)^2 + \cdots + r_N(x-a)^N$$

Using the Taylor-series expansion of a function

$$f(x) = f(a) + (x-a)f'(a) + (x-a)^2 \frac{f''(a)}{2!} + \cdots$$

we see that

$$\begin{aligned} r_0 &= P(a) \\ r_1 &= P'(a) \\ r_2 &= \frac{P''(a)}{2!} \\ r_3 &= \frac{P'''(a)}{3!} \\ &\cdots\cdots\cdots\cdots \end{aligned}$$

Using our earlier example,

$$\begin{array}{r|rrrrl}
2 & 1 & 0 & 6 & -7 & 8 \\
 & & 2 & 4 & 20 & 26 \\
\hline
2 & 1 & 2 & 10 & 13 & \boxed{34} = P(2) \\
 & & 2 & 8 & 36 & \\
\hline
2 & 1 & 4 & 18 & \boxed{49} = P'(2) & \\
 & & 2 & 12 & & \\
\hline
2 & 1 & 6 & \boxed{30} = \dfrac{P''(2)}{2} & & \\
 & & 2 & & & \\
\hline
 & 1 & \boxed{8} = \dfrac{P'''(2)}{6} & & &
\end{array}$$

and we have

$$P(x) = 34 + 49(x-2) + 30(x-2)^2 + 8(x-2)^3 + (x-2)^4$$

3.8 ROUNDOFF EFFECTS

The preceding is a purely mathematical result. How does it work out in the machine's floating-point number system with its ever-present roundoff? In particular, how large a remainder R can be considered as being zero?

We start by assuming that the coefficients a_i of the polynomial have roundoff errors in the floating-point form

$$a_i(1 + \varepsilon_i)$$

and we shall assume that the number a at which we want to evaluate the polynomial is exactly known. Since double-precision arithmetic is very widely available and usually costs little more to use than single precision,[1] it is widely used in polynomial evaluation. In this case the roundoff of the evaluation process contributes almost nothing to the error in the result—almost all the error arises from the initial coefficients. Since the coefficients occur linearly in the division process, the result is the sum of two divisions, one on the true coefficients a_i and the other on the errors $a_i \varepsilon_i$. If we assume that the ε_i are bounded,

$$|\varepsilon_i| \leq \varepsilon$$

then carrying out the following division process

$$\begin{array}{c|llll}
|a| & |a_N| & |a_{N-1}| & \cdots & |a_0| \\
\hline
 & |a_N| & \ldots\ldots\ldots\ldots & & \underline{|R} \\
\end{array}$$

gives $R\varepsilon$ as the error bound on the remainder. This is an error bound due to the error in the initial coefficients.

If single-precision arithmetic is used in the evaluation process, then we must also consider the effects of the rounding of the individual products and sums. Each arithmetic operation will include a factor

$$1 + \varepsilon^*$$

If we examine the process of synthetic division, we observe that the first coefficient a_N goes through $2N$ arithmetic operations and so picks up $2N$ factors of this form. The next coefficient a_{N-1} will have $2N - 1$ operations done on it, the next a_{N-2} will have $2N - 3$ operations, etc., down to an a_0 which will have 1 operation.

We are going to use the method of "backward analysis," meaning that the roundoff errors will be regarded as equivalent to small changes in the initial coefficients—we will answer the question "what problem has exactly the calculated answer?" The multiplication produces a factor $1 + \varepsilon$ on each of the terms that enter into the multiplicand. But the addition process may shift one number with respect to the other before the addition and roundoff take place. We are supplying a term $1 + \varepsilon$ to both numbers, thus putting in more roundoff than there should be, but the more the relative shift done, the less this excess. The typical term is of the form

$$a_k(1 + \varepsilon_k)a^k(1 + \varepsilon_1^*) \cdots (1 + \varepsilon_{2k+1}^*)$$

and the error in the polynomial evaluation from this term will be

$$a_k a^k[(1 + \varepsilon_k)(1 + \varepsilon_1^*) \cdots (1 + \varepsilon_{2k+1}^*) - 1]$$

[1] This is not true for the special-function evaluation of e^x, $\ln x$, $\sin x$, etc.

which is, neglecting products of ε_i's in comparison to ε_i's

$$a_k a^k[\varepsilon_k + \varepsilon_1^* + \varepsilon_2^* + \cdots + \varepsilon_{2k+1}^*]$$

For the purpose of bounding the ultimate error, all the ε^* are

$$|\varepsilon_j^*| \leq (\tfrac{1}{2})2^{-k}$$

for a k-bit machine. Thus if in place of a_k in the polynomial we wrote

$$a_k\left[1 + \varepsilon_k + \left(\frac{2k+1}{2}\right)2^{-k}\right]$$

we would have more than allowed for all the roundoff of the single-precision arithmetic done in the evaluation process, and we could proceed as before, using these modified coefficients to obtain a bound.

3.9 COMPLEX NUMBERS—QUADRATIC FACTORS

It frequently happens that a function with real coefficients is to be evaluated for a complex argument $x + iy$. For example, given

$$w = f(z) = \frac{e^z + e^{-z}}{2}$$

evaluate this for $z = x + iy$. We have

$$\begin{aligned} f(z) &= \frac{e^x \cos y + e^{-x} \cos y}{2} + \frac{ie^x \sin y - e^{-x} \sin y}{2} \\ &= \frac{e^x + e^{-x}}{2} \cos y + i\left(\frac{e^x - e^{-x}}{2}\right)\sin y \end{aligned}$$

It is also very common to be asked to find

$$f(z) + f(\bar{z})$$

where the bar means the conjugate $x - iy$. In the above example we will have

$$f(z) + f(\bar{z}) = (e^x + e^{-x})\cos y$$

The particular case of a polynomial is of importance, both because it occurs frequently and because it has special properties. The direct evaluation using complex arithmetic is unnecessarily expensive in machine time. We use the fact that evaluating the polynomial at a point $a + ib$ is the same as finding the remainder when dividing by the corresponding linear factor. Noting that corresponding to the linear factor we can construct the real quadratic factor

$$[z - (a + ib)][z - (a - ib)] = z^2 - 2az + a^2 + b^2$$

we divide the polynomial by this quadratic factor (using the obvious extension of synthetic division to quadratic factors) to get the remainder $r_1 z + r_0$.

Consider the special case of computing $P(2-3i)$ where, as before,

$$P(z) = z^4 + 6z^2 - 7z + 8$$

The quadratic factor is

$$[z-(2-3i)][z-(2+3i)] = z^2 - 4z + 13$$

$$\begin{array}{rr|rrrrr}
4 & -13 & 1 & 0 & 6 & -7 & 8 \\
\hline
 & & & & -13 & -52 & -117 \\
 & & & 4 & 16 & +36 & \\
\hline
 & & 1 & 4 & +9 & -23 & -109 = \text{remainder} \\
 & & \multicolumn{3}{c}{\underbrace{\qquad\qquad}_{\text{quotient}}} & &
\end{array}$$

Thus

$$P(z) = (z^2 - 4z + 13)(z^2 + 4z + 9) + (-23z - 109)$$

At $z = 2 - 3i$ we have $z^2 - 4z + 13 = 0$

so that

$$\begin{aligned} P(2-3i) &= -23(2-3i) - 109 \\ &= -155 + 69i \end{aligned}$$

Note how much complex arithmetic we have avoided in this process. This is a very useful trick: we replace the evaluation of a polynomial at a complex number by a division process using only real numbers and at the last moment let the complex number enter (only linearly).

Error bounds follow along the lines of the linear-factor theory, and need not be developed here in detail.

PROBLEMS

Evaluate the given polynomial for the argument value given.

3.9.1 $P(x) = x^6 + 6x^5 + 15x^4 + 20x^3 + 15x^2 + 6x + 1$; for $x = -1 + i$.

3.9.2 $P(x) = x^4 - 2x^3 + 5x^2 + 4x + 7$; for $x = \dfrac{-1 + i\sqrt{3}}{2}$.

3.9.3 $P(x) = x^4 - 2x^3 + 5x^2 + 4x + 7$; for $x = \dfrac{1 + \sqrt{5}}{2}$.

3.9.4 Discuss the evaluation of $P'(x + iy)$.

3.10 REPEATED EVALUATIONS

Often we want to evaluate a function not once, but at a series of equally spaced points (arguments). If machine time and costs are important, then significant savings can be made by taking advantage of various features of the function.

EXAMPLES

$$e^{a+(n+1)h} = e^{a+nh}e^h$$

$$\begin{cases} \sin[a+(n+1)h] = \sin(a+nh)\cos h + \cos(a+nh)\sin h \\ \cos[a+(n+1)h] = \cos(a+nh)\cos h - \sin(a+nh)\sin h \end{cases}$$

$$\ln[a+(n+1)h] = \ln(a+nh) + \frac{h}{a+nh} - \frac{1}{2}\left(\frac{h}{a+nh}\right)^2 + \cdots \qquad h \ll a+nh$$

////

For polynomials see Sec. 9.5.

The need for frequent evaluations of a polynomial even when the arguments are not equally spaced is a common occurrence. For example, most library routines for the special functions have some polynomial-evaluation steps. It comes as a surprise to many people to learn that for polynomials of degree six or higher there are arrangements that can save in the number of arithmetic operations used versus those used by the obvious chain method (see Ralston [49], sec. 7.2, for more details). Note that (1) often these shorter methods lead to severe roundoff troubles, and (2) the conversion to the shorter form can be programmed on a machine once and for all.

PROBLEMS

3.10.1 Discuss using the addition formula for $\tan x$ for finding successive, equally spaced values of $\tan x$. Include a discussion of the probable error trouble.

3.10.2 Note that the routine for successive values of the sine requires the corresponding cosine values; develop a method that uses only the last two sine values.

3.10.3 Examine the roundoff error propagation in the sine routine and the effect of writing $\cos h$ as $1 - \phi(h)$ for h small.

3.11 OVERFLOW AND UNDERFLOW

We have ignored one of the realities of computing, namely that not only are the mantissas limited, but so also are the exponents. This limitation gives rise to both overflow and underflow. It may seem strange at first that with the usual wide range of at least 10^{-38} to 10^{38} there should be trouble with exponents. But this ignores a number of facts. First, it is quite common to have adequate theories, with analytical results, at both ends of some range, and the computation is to cover the ground between where certain effects (terms) are completely negligible to where certain other effects are similarly to be ignored. Thus it is natural that in some cases being run there will be terms that are very small or very large.

Secondly, even for modest ranges some very common functions such as the exponential will have very large (small) values. For example, $e^{25} = 0.720 \times 10^{11}$.

Thirdly, many mathematical theories about the real world include singularities, and it is sometimes necessary to compute very close to the singularity in order to discover the nature of it. (See Chap. 42.)

When underflow occurs, it is almost always satisfactory to replace the number by the machine zero, 0.000×10^{-9}. Usually there is no comparable infinity to use when overflow occurs, and at best the overflow number is replaced by some very large number such as 0.900×10^9 (to leave room for some further increase without again tripping the overflow alarm).

Because of the lack of symmetry in the machine hardware and corresponding number system, it is preferable to arrange computations to have an underflow rather than an overflow. Usually this is not hard to do. For example, to compute

$$\frac{e^x}{e^x - 1} \qquad \text{for } 1 \leq x \leq 10$$

we write

$$\frac{e^x}{e^x - 1} = \frac{1}{1 - e^{-x}}$$

It should be obvious how to do this in many cases; what is necessary is to think about it *before* programming the details for a machine.

4

REAL ZEROS

4.1 INTRODUCTION

The problem of finding the real zeros of a continuous function occurs frequently in science and engineering and has therefore received extensive treatment. We shall give only a few of the simpler methods that are easy to understand and use. We will look at the problem of finding complex zeros in Chap. 5, and in Chap. 6 we will examine the important special case of polynomials.

It is in the nature of the problem of finding the real zeros of a function that the positive and negative terms of the function almost exactly cancel. We therefore face roundoff, and all that was said in Chap. 3 about how to evaluate a function to avoid unnecessary roundoff clearly applies to the problem of finding real zeros of a function.

The floating-point number system puts further restraints on what can be expected. We can at best hope to keep the relative error under control, and we cannot expect to find zeros far from the origin with great absolute accuracy. The roundoff also means that we cannot expect to find a number that makes the function exactly zero. It is for this reason that generally we do not try to find a zero; rather we try to find an interval in which the function changes sign. We then measure the accuracy by the length of the interval relative to the maximum of the function value and the argument. But even worse, we can expect in practice to find a sequence of values, some with plus and some with minus signs!

4.2 GRAPHICAL SOLUTION

One of the easiest methods for humans to use is the graphical method of drawing a curve and noting where it crosses the axis, or sometimes a pair of curves whose mutual crossings indicate the zeros. Unfortunately this method is particularly bad for unaided machines, and so we will look at it only briefly to set the stage for further methods that are more suitable for machines. For machines with graphical output and with a human examining the output, it can be very useful and easy to use.

EXAMPLE 4.1 Consider the problem of finding the real zeros of the function

$$y = e^{ax} - x^2$$

We picture in our minds the plot of $y = e^{ax}$ (see Fig. 4.2.1) and the plot of $y = x^2$ and look for their crossings. If a were small and positive, it is clear that the x^2 curve will rise quickly enough to cross the exponential curve at least once, and since ultimately the exponential grows faster than any power of x, there will be a second positive crossing.

When we plot the curves, we get a picture that confirms what we expected. It is natural to ask for the value of a beyond which there is only the negative zero and no crossing on the positive side. To find this we note that as a increases, the two positive crossings will approach each other and finally merge into a double zero. At this point both the function and the derivative will have a zero at the same place.

$$y = e^{ax} - x^2 = 0$$
$$y' = ae^{ax} - 2x = 0$$

Eliminate e^{ax} to get

$$ax^2 - 2x = 0$$

or

$$\begin{cases} x = 0 & \text{(spurious solution)} \\ x = \dfrac{2}{a} & \end{cases}$$

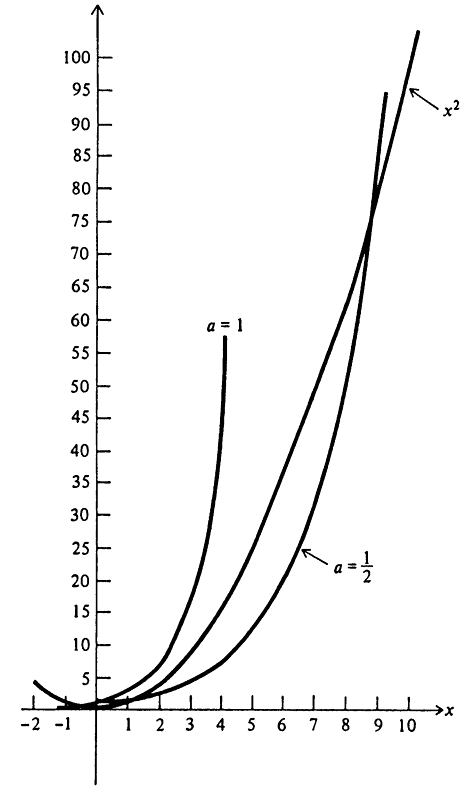


FIGURE 4.2.1

Putting this in the second equation, we get

$$ae^2 - \frac{4}{a} = 0$$

or for $a > 0$,

$$\begin{cases} a = \dfrac{2}{e} \\ x = e \\ y = 0 \end{cases}$$ ////

PROBLEMS

4.2.1 Find graphically the first two positive zeros of $\tan x - x = 0$ to one decimal place.
4.2.2 Find graphically the maximum of $y = -x \ln x (0 \leq x \leq 1)$. Check analytically.
4.2.3 Find the value of a for which $\sin x - ax = 0$ has a double zero.
4.2.4 Show that there are no real zeros for $y = e^x - \ln x - 1$.

4.3 THE BISECTION METHOD

One of the best, most effective methods for finding the real zeros of a continuous function is the bisection method. The bisection method begins with an interval (x_1, x_2) in which the function changes sign, which is measured in the machine by

$$f(x_1)f(x_2) < 0$$

We then pick the midpoint x_3 (bisect the interval) and evaluate the function there. If

$$f(x_1)f(x_3)\begin{cases} < 0 & \text{then there is a sign change in } (x_1, x_3) \\ > 0 & \text{then there is a sign change in } (x_3, x_2) \\ = 0 & \text{then } x_3 \text{ is a zero} \end{cases}$$

Repeated bisections reduce the interval containing the sign change to an arbitrarily small length—or until we meet the granularity of the number system around x and the machine computes the midpoint as one of the two end values. Ten bisections reduce the interval length by more than a factor of $\frac{1}{1{,}000}$—three decimal place improvement! Thus in practice it is rare that the bisection method will be used for 20 steps.

The method is *robust* in the sense that small roundoff errors will not prevent the method from giving an interval with a sign change, and if roundoff is misleading you, it is not the fault of the method but of the program that evaluates the function. There is one danger, namely that what you implicitly thought was a continuous function had a pole and that you end up straddling an odd-order pole, but this can be checked for easily.

The bisection method supposed that an interval had been found which had a sign change in it, and the problem is usually stated as that of finding all the real zeros in some given interval. This requires a search for intervals of sign change. If we search with a large step size h, then we run the risk of stepping over a pair of zeros or of getting an interval with an odd number of zeros and finding only one of them in the end. If we use a small step size h, then we will spend most of the time looking where there are no zeros. *It is an engineering judgment to resolve this dilemma.*

The problem of multiple zeros can be troublesome. One way to cope with it is to search for sign changes in both the function (which indicates all odd-order zeros) and the first derivative (which indicates all even-order zeros), but without a lot of elaborate analysis the machine cannot be expected to solve the general problem of the multiplicity of the zeros it isolates.

EXAMPLE 4.2 Find $\sqrt{2}$.

This is a zero of $y = x^2 - 2$. Try $h = \frac{1}{2}$, starting at $x = 0$, for the search.

$$y(0) = -2$$
$$y(\tfrac{1}{2}) = -\tfrac{7}{4}$$
$$y(1) = -1$$
$$y(\tfrac{3}{2}) = \tfrac{1}{4}$$

We have the interval

$$1 < x < 1.5$$

Bisect and try $x = 1.25$; $y(1.25) = -0.4375$. We have the interval

$$1.25 < x < 1.5$$

Bisect and try $x = 1.37$; $y(1.37) = -0.1231$. We have the interval

$$1.37 < x < 1.5$$

Bisect and try $x = 1.43$; $y(1.43) = 0.0449\ldots > 0$. And so forth. ////

As a practical matter, given the original interval, it is usually best to decide on the *number* of subintervals to try in the search method and on the number of bisections to use in case a sign change is found, rather than on interval sizes.

PROBLEMS

Find to one decimal place the real zero of:

4.3.1 $he^h = 1$

4.3.2 $\cos x = x$

4.3.3 $x \ln x = 1$

Using the bisection method, compute:

4.3.4 $\dfrac{1 + \sqrt{5}}{2}$ *Hint:* Find a quadratic of which it is a zero.

4.3.5 $\sqrt[3]{2}$

4.3.6 Elaborate on the last paragraph of this section.

4.4 THE METHOD OF FALSE POSITION

The method of false position (*regula falsi*) is a very ancient method, and it attempts to do better than the obviously slow bisection method. The method of false position again starts with two points at which the function has opposite signs; that is, we have an interval in which there is a zero, and we wish to decrease the interval. The straight line through the two points is used in place of the function, and the zero of the straight line is used in place of the midpoint of the bisection method. The main weakness of the method—its slow approach from one side—is apparent from the picture. Further obvious faults tend to exclude the method from practical use on machines.

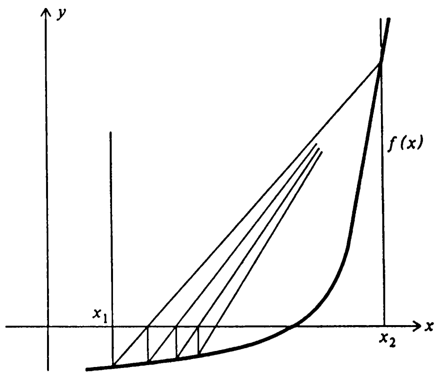


FIGURE 4.4.1

In more detail, given the two points $(x_1, f(x_1))$ and $(x_2, f(x_2))$ (see Fig. 4.4.1), the line through them is

$$y(x) = y(x_1) + \frac{y(x_2) - y(x_1)}{x_2 - x_1}(x - x_1)$$

whose zero is given by

$$\begin{aligned} x &= x_1 - y(x_1)\frac{x_2 - x_1}{y(x_2) - y(x_1)} \\ &= \frac{x_1 y(x_2) - x_2 y(x_1)}{y(x_2) - y(x_1)} \end{aligned}$$

Note (1) the symmetry of the formula and (2) that $y(x_2) - y(x_1)$ is in fact an addition because we assumed $y(x_1)y(x_2) < 0$.

4.5 MODIFIED FALSE POSITION

A simple modification of the false-position method eliminates its worst feature, the one-sided approach to the zero with the resulting large interval. The modification of the false-position method consists in dividing by 2 during each cycle of the computation the function value at the end that is kept. Figure 4.5.1 shows how this modification improves the method.

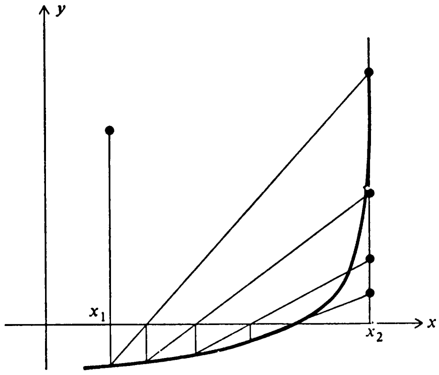


FIGURE 4.5.1

The method is usually (but not always) faster than the bisection method and has the defect that we cannot tell in advance how many steps will be required to obtain an interval length less than the preassigned accuracy needed.

In more detail the algorithm is the following. Given a continuous function $f(x)$ for an interval $a \leq x \leq b$ with $f(a)f(b) < 0$,

1 Compute $x = \dfrac{af(b) - bf(a)}{f(b) - f(a)}$.

2 Compute $f(x)$.

3 Compute $f(a)f(x)$. If $f(a)f(x)$

$$\begin{cases} < 0 & \text{then} \begin{cases} x & \text{becomes new } b \\ f(x) & \text{becomes new } f(b) \\ \dfrac{f(a)}{2} & \text{becomes new } f(a) \end{cases} \\ = 0 & \text{then you are done} \\ > 0 & \text{then} \begin{cases} x & \text{becomes new } a \\ f(x) & \text{becomes new } f(a) \\ \dfrac{f(b)}{2} & \text{becomes new } f(b) \end{cases} \end{cases}$$

EXAMPLE 4.3 Find the zero of $y = he^h - 1$ (using two-decimal arithmetic to make it easily followed).

It is easy to see that

$$f(0) = -1 < 0$$
$$f(1) = e - 1 > 0$$

hence

$$a = 0 \qquad f(a) = -1$$
$$b = 1 \qquad f(b) = 1.72$$
$$x = \frac{af(b) - bf(a)}{f(b) - f(a)} = \frac{0 \times 1.72 - 1(-1)}{1.72 - (-1)} = \frac{1}{2.72} = 0.37$$
$$f(x) = 0.37 \times 1.45 - 1 = 0.54 - 1 = -0.46$$

next step

$$a = 0.37 \qquad f(a) = -0.46$$
$$b = 1 \qquad \frac{f(b)}{2} = 0.86$$
$$x = \frac{0.37(0.86) - 1(-0.46)}{0.86 + 0.46} = \frac{0.778}{1.32} = 0.59$$
$$f(x) = 0.59 \times 1.80 - 1 = 0.06$$

next step

$$a = 0.37 \qquad \frac{f(a)}{2} = -0.23$$
$$b = 0.59 \qquad f(b) = 0.06$$
$$x = \frac{0.37(0.06) - 0.59(-0.23)}{0.06 + 0.23} = \frac{0.16}{0.25} = 0.55$$
$$f(x) = 0.55 \times 1.73 - 1 = -0.05$$

next step

$$a = 0.55 \qquad f(a) = -0.05$$

$$b = 0.59 \qquad \frac{f(b)}{2} = 0.03$$

$$x = \frac{0.55(0.03) - 0.59(0.05)}{0.03 + 0.05} = \frac{0.046}{0.08} = 0.58$$

$$f(x) = 0.038$$

$$a = 0.55 \qquad \frac{f(a)}{2} = -0.025$$

$$b = 0.58 \qquad \frac{f(b)}{2} = 0.038$$

$$x \approx 0.565$$

End. ////

An alternate version of the modified false-position method halves the ordinate kept *only* when it stays on the same side of the zeros as on the previous step. This modification is believed to speed up the convergence at the cost of a slight amount of extra programming and testing. Unfortunately the method has not been adequately analyzed, and experimental tests of selected functions are not completely convincing.

PROBLEMS

Using the modified false-position method for three steps, find the zero of:

4.5.1 $y = \tan x - \dfrac{1}{1 + x^2} \qquad 0 \leq x < \dfrac{\pi}{2}$

4.5.2 $y = x^2 - 2 \qquad 0 \leq x \leq 2$

4.5.3 $y = x \ln x - 1 \qquad 1 \leq x \leq 2$

4.5.4 $y = \cos x - x \qquad 0 \leq x \leq \dfrac{\pi}{2}$

4.5.5 $y = x^3 - 2 \qquad 0 \leq x \leq 2$

4.5.6 The choice of halving the kept function value was arbitrary. Discuss other possible choices and when to use them.

4.5.7 Draw a flow diagram of the modified false-position method.

4.6 NEWTON'S METHOD

Newton's method for finding the real zeros of a function $y = f(x)$ is usually taught in the calculus course. The idea behind the method is the "analytic substitution" of the local tangent line for the function and then the use of the zero of this line as the next approximation to the zero of the function.

In mathematical notation the tangent line at x_k (Fig. 4.6.1) is

$$y(x) = f(x_k) + f'(x_k)(x - x_k)$$

and hence solving for $x = x_{k+1}$, the next approximation is

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

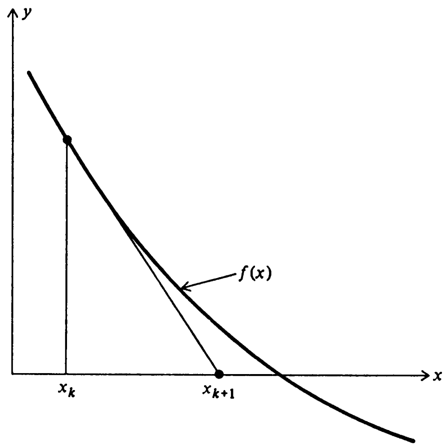
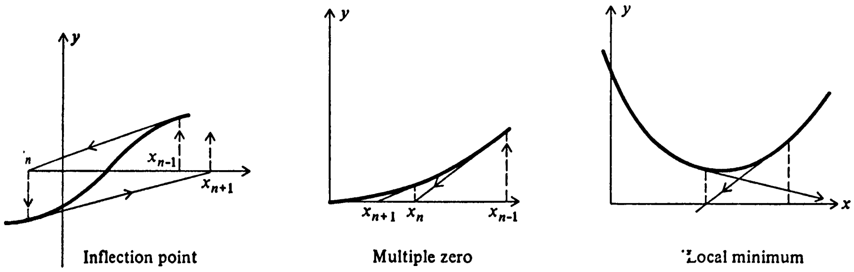


FIGURE 4.6.1

When it works, Newton's method is fine, but it has three obvious faults as can be seen in Fig. 4.6.2. Thus, unless the local structure of the function is known in some detail, the unmodified Newton's method is to be avoided.

The worst features of the method can be partially compensated for by a simple device. Let us introduce a *metric*, or distance, to measure how well we are doing. We will use $|f(x)|$ as this metric. The beginning of the next step in the iteration process involves the evaluation of $f(x)$ at the new point. If this new



Can cycle and never converge

Slow approach with $f' \to 0$ and trouble in division step

Risks being sent very far away for next approximation

FIGURE 4.6.2

value is not smaller in size than the previous value, we will not accept the step, but instead we will go back and halve the step

$$x_{k+1} - x_k = \Delta x_k = \frac{-f(x_k)}{f'(x_k)}$$

using instead

$$\frac{\Delta x_k}{2}$$

If this does not decrease the distance, we repeat the halving process until it does, because we *know* that the local slope of the curve is in the direction to decrease the distance and that a sufficiently small step will work (until the granularity of the number system gets in the way). If this modification is used, it is well to add another, namely that no new step is more than twice the preceding step (to prevent the tedious shortening-up of the step size cycle after cycle). In this way the troubles from inflection points are eliminated, and the method will creep up on multiple zeros. Local minima, which are not always easily recognized, can cause trouble in a machine.

How shall we end the iteration in Newton's method? The approach to the zero becomes one-sided, and we do not have an interval in which the zero lies. If we use the size of the function at the point as the criterion, we will get different answers depending on whether we use $f(x) = 0$ or $kf(x) = 0$ as our original equation. If we use the step size $\Delta x = x_{k+1} - x_k$ as a measure, then when we quit, we may be far from the zero and be only slowly creeping up on it. There is no completely satisfactory answer, and we will suggest using the size of the step *relative* to the zero being found as a measure. As will be shown in the Sec. 4.7, the error tends to be squared each step when we are finally close (in some sense), and this fact can be used by comparing three consecutive step sizes,

$$\frac{\Delta x_{k+1}}{(\Delta x_k)^2} \approx \frac{\Delta x_k}{(\Delta x_{k-1})^2} \approx \frac{-f''(x_{k+1})}{2f'(x_{k+1})}$$

4.7 THE CONVERGENCE OF NEWTON'S METHOD

Newton's method is highly favored theoretically because when it finally starts to converge, the convergence is quite rapid. To show this we need a couple of important mathematical results.

The first result we need is the Taylor series with an integral remainder. For our immediate use this can be found by integrating by parts the obvious identity

$$y(x) = y(a) + \int_a^x y'(s)\, ds$$

Setting

$$u = y'(s) \qquad du = y''(s)\, ds$$
$$dv = ds \qquad v = -(x - s)$$

we have

$$\begin{aligned} y(x) &= y(a) - [(x-s)y'(s)]\Big|_a^x + \int_a^x y''(s)(x-s)\, ds \\ &= y(a) + (x-a)y'(a) + \int_a^x y''(s)(x-s)\, ds \end{aligned}$$

The second mathematical result we need is that if $f(x)$ and $g(x)$ are continuous and if $f(x) \geq 0$ $(a \leq x \leq b)$, then

$$\int_a^b f(x)g(x)\, dx = g(\theta) \int_a^b f(x)\, dx \qquad \text{for some } \theta \qquad a < \theta < b$$

To prove this, write g as the minimum of $g(x)$ and write G as the maximum (in the interval). Thus

$$g \leq g(x) \leq G$$

Multiply this inequality by the nonnegative quantity $f(x)$

$$gf(x) \leq f(x)g(x) \leq Gf(x)$$

and then integrate (sum) to get

$$g \int_a^b f(x)\, dx \leq \int_a^b f(x)g(x)\, dx \leq G \int_a^b f(x)\, dx$$

Now consider the quantity, as a function of t,

$$\phi(t) = g(t)\int_a^b f(x)\,dx - \int_a^b f(x)g(x)\,dx$$

At the point where

$$g(t) = g, \text{ then } \phi(t) \le 0$$
$$g(t) = G, \text{ then } \phi(t) \ge 0$$

Hence, since $\phi(t)$ is continuous, there is a value $t = \theta$ such that

$$\phi(\theta) = 0$$

or

$$\int_a^b f(x)g(x)\,dx = g(\theta)\int_a^b f(x)\,dx$$

We use these results as follows: write $f(x)$ in the Taylor-series form where a is the current guess x_k and x is the zero of the function.

$$f(x) = 0 = f(x_k) + (x - x_k)f'(x_k) + \int_{x_k}^{x} (x - s)f''(s)\,ds$$

The next guess x_{k+1} is given by Newton's formula

$$0 = f(x_k) + (x_{k+1} - x_k)f'(x_k)$$

Subtracting, we get

$$(x - x_{k+1})f'(x_k) + \int_{x_k}^{x} (x - s)f''(s)\,ds = 0$$

Using the second theorem, since $x - s$ is of constant sign,

$$(x - x_{k+1})f'(x_k) + f''(\theta)\int_{x_k}^{x} (x - s)\,ds = 0$$

$$(x - x_{k+1})f'(x_k) + f''(\theta)\frac{(x - x_k)^2}{2} = 0$$

But

$$x - x_{k+1} = \varepsilon_{k+1} = \text{next error}$$
$$x - x_k = \varepsilon_k = \text{current error}$$

hence

$$\varepsilon_{k+1} = \frac{-f''(\theta)}{2f'(x_k)}\,\varepsilon_k^{\,2}$$

is an exact formula for the new error in terms of the current error; we square the error each step (almost). This is often described by the remark that with Newton's method you double the number of correct digits each step (assuming that $f''(0)/[2f'(x_k)]$ is around 1 in size).

But let us not be deceived by this result. Normally it is not the final rate of convergence that controls the number of iterations; it is the initial rate of convergence. Here the bisection method shines—it starts out well in comparison with many other methods.

PROBLEMS

Apply Newton's method, using the starting value given to:

4.7.1 $y = xe^x - 1 \qquad x = 1/2$

4.7.2 $y = \arctan x - 1 \qquad x = 1$

4.7.3 $y = \ln x - 3 \qquad x = 10$

4.7.4 Using $y = x^n - 1$, show how for some values of x_k the local convergence of Newton's method can be very slow.

4.8 INVARIANT ALGORITHMS

The idea of an invariant algorithm is simple but important. The importance arises from being one of the few ideas that provide some unity in the field of algorithms with its many varied methods and tricks.

The idea is easily illustrated by applying it to Newton's method. We instinctively feel that it is the same problem whether we find the zeros of $y = f(x)$ or of $y = cf(x)$. A moment's examination of Newton's method shows that at the step where we compute $f(x)/f'(x)$ the multiplicative factor c is automatically removed. We also feel that if in place of x we were to use $c_1 x$ as the argument of the function, then we still have essentially the same problem. *If* we scale the starting value by the corresponding amount, then we again see that the successive values of the estimates x_k also scale properly; the iteration equation is homogeneous in c_1. Due to the nature of the floating-point number system a translation of the origin *does not* leave the problem unchanged.

What about the stopping rule? Should it not also be invariant under these two classes of transformations? The use of the change in the estimate $x_{k+1} - x_k$ *relative* to the estimate x_k clearly scales properly. The use of the step size, the use of the size of the function, and many other possible stopping rules do not scale properly and are not appropriate for an invariant algorithm.

In general, an algorithm is invariant with respect to some class of transformations. The class is found by considering the source of the problem, the mathematical structure of the problem, and the use that is going to be made of the results.

The class usually forms a *group*, provided the finite limitations of the machine are recognized. The importance of invariance with respect to a group of transformations has long been recognized in mathematics. That which is invariant is usually recognized as being more fundamental than the chance form of the particular representation of the problem; in Euclidean geometry only things invariant with respect to translation and rotation are regarded as geometric entities. Similarly, an algorithm which transforms properly with respect to a class of transformations is more basic than one that does not. In a sense the invariant algorithm attacks the problem and not the particular representation used (though clearly it uses a particular representation in the computation done). The slight differences in roundoff that can occur are not basic to the idea, but come from the nature of the finite structure of the number system used. On the other hand, the roundoff differences from different representations will *usually* not be large, except for ill-posed problems.

Invariance is something like dimensional analysis. We realize instinctively that invariance should apply to proper equations and proper algorithms, and if it does not, then this is a warning sign to look a good deal closer before going on. And like dimensional analysis, invariance in an algorithm can point to forms that are suitable and ones that are not, and often delimit the acceptable forms so as to practically give the proper one (see, for example, the stopping rule for Newton's method). But invariance may include transformations that are more than mere linear scaling, and so the idea of invariant algorithms includes more than dimensional analysis.

The larger the class of transformations under which the algorithm is to be invariant, the more restricted the algorithm and (like much of mathematics) the more simple and powerful the result when finally found. In the search for an algorithm the invariance requirement will block off many fruitless paths.

It is easy to see that two classes of transformations, multiplying the function by a constant and stretching the x axis, are reasonable to be used on the problem of finding zeros of functions. Applying this invariance to the bisection method, we see that if the initial interval is properly scaled, then it is a matter of scaling the stopping rule; and if the stopping rule is not so scaled, then there is probably something wrong.

Similarly the false-position and modified false-position methods are invariant *provided* the stopping rules are picked properly. In particular, the method of iterating the bisection method a fixed number of times, as well as the interval length relative to the size of the zero being found, is invariant. Stopping rules, like the size of the function or the length of the interval, do not scale properly and should be avoided in practice.

In the future we will use this idea of invariance as a tool for finding and examining algorithms, and we will find that some of the classic methods are not invariant.

4.9 REMARKS ON COMPARING ALGORITHMS

Having given a number of algorithms for finding real zeros, how are we to compare them and others yet to be discovered so that we can pick one in a reasonable fashion when necessary to do a particular problem?

Evidently the bisection method is powerful, but slow in the final steps when compared to Newton's method. But it is quite likely that the bisection method is much faster in the early stages. And in practice it is usually the initial rates of convergence, not the final ones, that matter because we rarely want high accuracy in the zeros we find—in mathematical problems yes, but in engineering and science usually we don't know the other parts sufficiently well to justify a many-digit answer.

The modified false position is so much better than the simple false position that it is the second choice to consider. When and in what respects does it compare favorably with the bisection method? Evidently it is likely to be faster on reasonable functions, but it can be slower on some functions. Nothing is really known at this time about the distribution of functions that occur in practice whose zeros we want to find, and so little can be said.

Newton's method is the classic one for finding zeros. It requires finding and coding the derivative. Some methods of rating algorithms count the effort to evaluate the derivative the same as to evaluate the function—thus counting one step as the equal of two in other methods—but this is *not* reasonable. For most functions, once the parts of the function have been found, there are very few new parts that are expensive to compute when the derivative is found. No new radicals, logs, or exponentials can arise; at worst a sine can go into a cosine (or the reverse), and the special functions are the time-consuming parts of most function evaluations. All the straight arithmetic is usually small in time when compared to the special-function-evaluation routines..

Two methods which are sometimes discussed, but which we have so far ignored, should be mentioned. The *secant method* resembles the false-position method, *except* that it uses the plausible argument that the last two values, rather than a pair that lies one on each side, should be kept. The danger in the method is especially clear when roundoff is considered, since by chance a locally horizontal tangent can cause great trouble.

The other method is called *successive substitutions.* Given one or more equations, they are each solved explicitly for a different variable, even if it also appears on the other side, and the old values are substituted into one side to find updated new values of the unknowns. This method is powerful when done by hand, but it is hard to use on machines in situations that have not been preanalyzed. Evidently its success depends on arranging matters so that large errors in an unknown tend to produce smaller ones in the next cycle of iteration. This technique is related to the idea of stability which will be discussed in more detail in later chapters.

EXAMPLE 4.4 Consider again the example in Sec. 4.2 of finding the zeros of

$$e^{x/2} - x^2 = 0$$

this time using $a = \frac{1}{2}$.
We have the two curves (Fig. 4.9.1)

$$y = x^2$$
$$y = e^{x/2}$$

Solving the second equation for x, we have

$$y = x^2$$
$$x = 2 \ln y$$

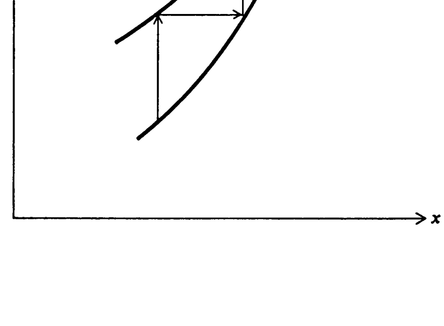


FIGURE 4.9.1
Rough Sketch.

If we start with $x = 5$, we get in turn (using two decimals)

$$\begin{aligned}
x &= 5 \\
y &= x^2 = 25 \\
x &= 2 \ln y = 6.4 \\
y &= 41 \\
x &= 7.4 \\
y &= 55 \\
x &= 8.0 \\
y &= 64 \\
x &= 8.3 \\
y &= 69 \\
x &= 8.5 \\
y &= 72 \\
x &= 8.5
\end{aligned}$$

(which is a poor answer), and we approach the upper zero from below. If we were to reverse the substitution loop, we would approach the lower zero from above. ////

It is easy to draw sketches to show what will happen in particular cases, but this is exactly the kind of thing that machines have a lot of trouble doing. Thus the successive-substitution method is suitable only for carefully analyzed situations (of which there are many), but the details are easily worked out in any particular situation, and nothing more need be said here.

PROBLEMS

4.9.1 Make sketches showing the dangers of the secant method.
4.9.2 Show that the proposed modification of Newton's method is invariant.

4.10 TRACKING ZEROS

The problem is often not just to find the real zeros of a function, but rather to find the zeros as a function of some parameter in the equation. In this situation it is clearly foolish to solve each parameter value as if it were a new problem. Instead, from the values of the zeros for the previous parameter value(s) we can form a good guess at where those for the current parameter value probably lie. Methods for tracking the zeros as functions of the parameter can be developed from the predictor-corrector methods discussed in Chap. 23. It is a complex, difficult task to design a foolproof method of tracking zeros since sooner or later almost every possible trouble will occur.

In tracking zeros one should use a local coordinate system (like the moving trihedral in differential geometry) to avoid artifacts of the problem's coordinate system such as vertical and horizontal tangents. This remark applies to almost all graphical work.

For this chapter probably the best, easily available references are Ralston [49] and Traub [59] (see References in the back of this book).

5

COMPLEX ZEROS

5.1 INTRODUCTION

The problem of finding the complex zeros of a function in some region occurs frequently in practice, though not as frequently as finding real zeros. It is curious that the real-zero problem has been extensively investigated while the complex-zero problem has generally been ignored. Evidently it is a field ripe for further research. We shall give only two simple methods which are based to a great extent on the bisection method.

The functions we shall consider are called "analytic in the region of investigation," meaning that at every point in the region the function has a convergent Taylor series. For convenience only, we shall assume the region is a rectangle in the complex plane.

While finding real zeros, we used the notation $y = f(x)$; in the complex plane it is customary to use the notation $z = x + iy$ as the independent variable and $w(z) = u(x, y) + iv(x, y)$ as the dependent variable. Thus the single condition that $w(z) = 0$ is equivalent to the two simultaneous conditions

$$u(x, y) = 0$$
$$v(x, y) = 0$$

where $u(x, y)$ is called *the real part* and $v(x, y)$ is called *the imaginary part* of $w(z)$.

Very frequently the function we are examining has only real values for $w(z)$ when z takes on real values; that is, $w(x + i0)$ has only real values. In this important case we have the well-known result that if $x + iy$ is a zero of $w(z)$, then $x - iy$ is also a zero.

The theorem is important because it means that in this very common case we need to examine only the upper (or lower) half of the complex plane for zeros.

As an exercise in dealing with complex numbers we shall prove this theorem. The letter i can occur in the function definition as well as in the argument $z = x + iy$; for example,

$$w(z) = \sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

and we need to distinguish these two appearances. Given $w = f(z)$, we shall use a long bar over both the function and the argument to mean the conjugate values, that is, when i is replaced by $-i$. A short bar over the function means that only the i's in the function are changed, while the bar over the argument means that only in the argument are the i's changed to $-i$'s. The assumption that the function takes on real values for real arguments means that, as in the above case of $\sin x$,

$$w = w(x) = \bar{w}(x)$$

The statement that $x + iy$ is a zero means that

$$w(x + iy) = 0$$

We need to show also that $w(x - iy) = 0$. We have in turn

$$w(x + iy) = 0 = \overline{w(x + iy)} = \bar{w}(x - iy) = w(x - iy)$$

and we have proved the theorem.

PROBLEMS

For practice in handling complex functions, find u and v

5.1.1 If $w = z^3$

5.1.2 If $w = \tan z$

5.1.3 If $w = \ln z$

5.1.4 If $w = \sqrt{z}$

5.2 THE CRUDE METHOD

The bisection method for finding real zeros is both very easy to understand and very robust, and it is natural to try to extend the method to finding complex zeros. This extension depends on finding the right way of looking at the bisection method in order to generalize it to complex zeros (keeping in mind the invariance principle).

One approach is to regard the search method for locating the zero as recording plus or minus at each point along the real axis according to the sign of the function at the point. In the complex plane for each point $x + iy$ we record

the quadrant number in which the function value falls. Thus at each point, we record 1, 2, 3, or 4, and we record a 0 whenever either $u(x, y) = 0$ or $v(x, y) = 0$ or both. This produces a picture of a region with numbers attached to each mesh point (see Fig. 5.2.1).

We can now take colored pencils and color in each quadrant. We will find that typically four quadrants meet at a point, each having an angle of about 90° at the point. At this point there is evidently a zero, since it corresponds to a point in the w plane where both $u(x, y)$ and $v(x, y)$ are zero.

The $u(x, y) = 0$ curves divide quadrants 1 and 2, and 3 and 4, while the $v(x, y) = 0$ curves divide quadrants 1 and 4, and 2 and 3.

As we shall later show (Sec. 5.4), where four quadrants meet at a point, it is a simple zero; where eight quadrants meet (each having an angle of about 45°), it is a double zero; etc.

Having located the general region of the zero, we may clearly enlarge the region as much as we please by plotting our mesh points at finer and finer spacing to get further accuracy until we run into either roundoff or the granularity of the number system.

This crude method is easy to understand, reliable, and accurate, but its fault is that it requires unnecessary computing (and also implies human intervention in the process of finding the zeros). We will later refine the method, but even in the crude form it is very useful and effective.

PROBLEMS

5.2.1 Prove that for functions such that $w = \bar{w}$, the x axis of the plot is a line of zeros corresponding to $v = 0$.

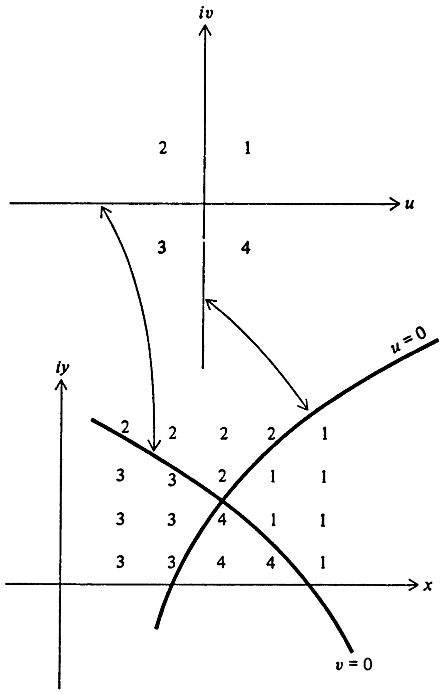


FIGURE 5.2.1

5.3 AN EXAMPLE USING THE CRUDE METHOD

As an example of how the crude method works, consider the problem of finding the complex zeros of the function

$$w = w(z) = e^z - z^2$$

which lie near the origin. For this function we have $w(x) = \bar{w}(x)$, and so we need only explore the upper half-plane. We try the rectangular region

$$-\pi \leq x \leq 2\pi$$
$$0 \leq y \leq 2\pi$$

The quadrant numbers are easily computed on a machine and are plotted in Fig. 5.3.1. The x axis is a line of 0s, as it should be (there is one other number which is almost zero which we have marked as a 0). In drawing the curves $u = 0$

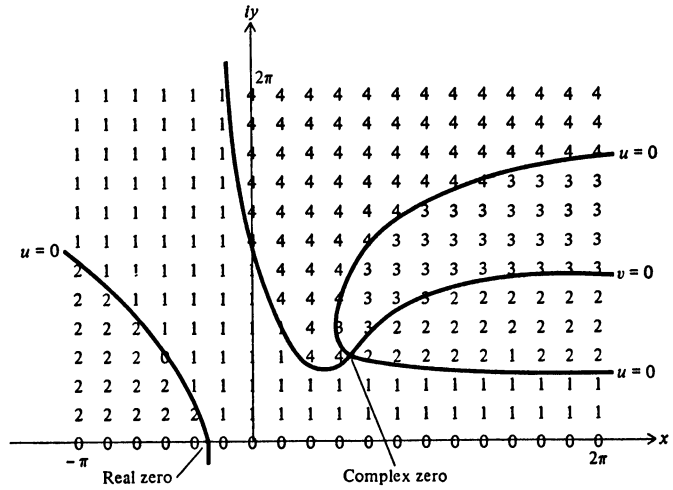


FIGURE 5.3.1

and $v = 0$, we remember that the curves in the lower half-plane are the mirror images (in the x axis) of those in the upper half, but that the quadrant designations interchange 1 and 4 as well as 2 and 3.

Does this picture seem to be reasonable? The real zero on the negative real axis seems to be about right because we know that as x goes from 0 through negative values, e^x decreases from 1 toward 0, whereas z^2 goes from 0 to large positive values. Thus, e^x and z^2 must be equal at some place (and we easily see that this happens before we reach -1).

The complex zero we found is likely to be one of a family of zeros with the next one appearing in the band

$$2\pi \leq y \leq 4\pi$$

An examination of the picture shows that it is a reasonably convincing display of the approximate location of the complex zero. We could easily refine the particular region if we wished by simply placing our points in a closer mesh.

5.4 THE CURVES $u = 0$ AND $v = 0$ AT A ZERO

At any point $z = z_0$ (Fig. 5.4.1) the Taylor expansion of a function has the form

$$f(z) = f(z_0) + f'(z_0)\frac{z - z_0}{1!} + f''(z_0)\frac{(z - z_0)^2}{2!} + f'''(z_0)\frac{(z - z_0)^3}{3!} + \cdots$$

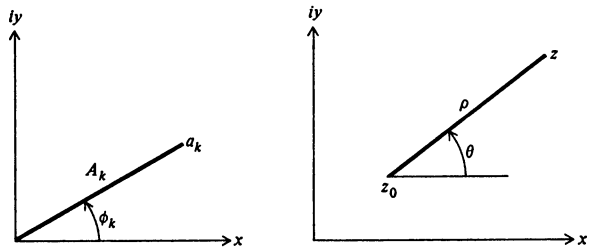


FIGURE 5.4.1

or

$$f(z) = a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + a_3(z - z_0)^3 + \cdots$$

For each k we set

$$a_k = A_k e^{i\phi_k} \qquad A_k \text{ real}$$

and we also set

$$z - z_0 = \rho e^{i\theta}$$

Therefore, the Taylor expansion has the form

$$f(z) = A_0 e^{i\phi_0} + A_1 \rho e^{i(\phi_1 + \theta)} + A_2 \rho^2 e^{i(\phi_2 + 2\theta)} + \cdots$$

At a simple zero $f(z_0) = 0$, then $A_0 = 0$, and for small ρ the immediate neighborhood of z_0 $f(z)$ "looks like"

$$A_1 \rho e^{i(\phi_i + \theta)}$$

or

$$f(z) \approx A_1 \rho[\cos(\phi_1 + \theta) + i \sin(\phi_1 + \theta)]$$

The $u = 0$ curves (Fig. 5.4.2) are approximately given by

$$A_1 \rho \cos(\phi_1 + \theta) = 0$$

or

$$\theta = -\phi_1 + \frac{\pi}{2} + k\pi \qquad k = 0, 1$$

and the $v = 0$ curves are approximately given by

$$A_1 \rho \sin(\phi_1 + \theta) = 0$$

or

$$\theta = -\phi_1 + k\pi \qquad k = 0, 1$$

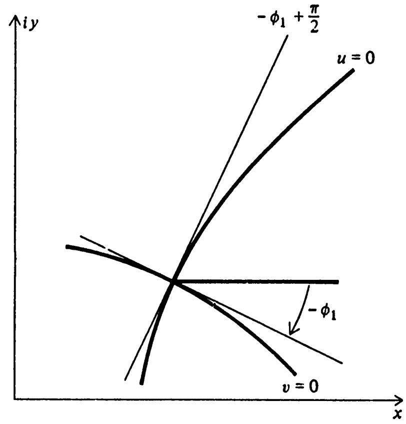


FIGURE 5.4.2

We see that the $u = 0$ and $v = 0$ curves intersect at right angles, and hence the quadrants we plan to color each have an angle of approximately 90° at the zero. The picture we color is therefore easy to interpret at a simple zero. Note that this happened at both zeros of the example in Sec. 5.3.

At a double zero both $f(z_0)$ and $f'(z_0)$ are 0, so that the Taylor series looks like

$$f(z) = A_2 \rho^2 e^{i(\phi_2 + 2\theta)} + A_3 \rho^3 e^{i(\phi_3 + 3\theta)} + \cdots$$

We have for small ρ

$$\begin{aligned} u &\approx A_2 \rho^2 \cos(\phi_2 + 2\theta) \\ v &\approx A_2 \rho^2 \sin(\phi_2 + 2\theta) \\ u = 0 \qquad \theta &= -\frac{\phi_2}{2} + \frac{\pi}{4} + \frac{k\pi}{2} \\ v = 0 \qquad \theta &= -\frac{\phi_2}{2} + \frac{k\pi}{2} \end{aligned}$$

and the angles of the colored quadrants are approximately 45° (see Fig. 5.4.3). It is easy to see that for a triple zero the colored quadrant angles will be approximately 30°, and in general for a multiplicity of m, we shall have the $u = 0$ and $v = 0$ curves meeting at approximately $\pi/(2m)$ radians (or $90°/m$).

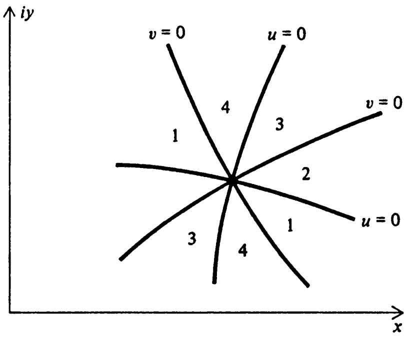


FIGURE 5.4.3

As an example of a double zero, consider Example 4.1 in Sec. 4.2 where we asked for a double zero of

$$w(z) = e^{az} - z^2$$

and found that $a = 2/e$. What does the picture of this look like? We should have the u and v curves crossing at 45° angles for this transcendental function. We have plotted the quadrant numbers only where quadrant changes occur, which is all that is needed. (See Fig. 5.4.4.)

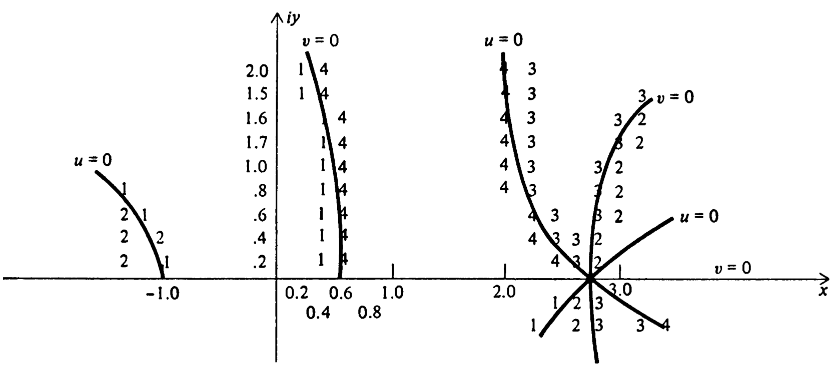


FIGURE 5.4.4

PROBLEMS

5.4.1 Sketch the $u = 0$ and $v = 0$ curves for $w = z^2 - x + \frac{1}{2}$ using the lattice points

$$x = \frac{k}{4} \qquad k = 0, 1, 2, 3, 4$$

$$y = \frac{m}{4} \qquad m = 0, 1, 2, 3, 4$$

and check by using the quadratic equation formula.

5.4.2 Prove that for a zero of order m, the quadrant angles are $\pi/(2m)$ radians.

5.5 A PAIR OF EXAMPLES OF $u = 0$ AND $v = 0$ CURVES

The following pair of simple polynomials illustrate how the $u = 0$ and $v = 0$ curves behave at zeros and elsewhere in the plane.

The first example is a simple cubic with zeros at -1, 0, and 1. The polynomial is

$$\begin{aligned} w = w(z) &= (z + 1)z(z - 1) = z^3 - z \\ &= (x + iy)^3 - (x + iy) \\ &= (x^3 - 3xy^2 - x) + i(3x^2y - y^3 - y) \end{aligned}$$

The real curves are defined by

$$u = x^3 - 3xy^2 - x = 0$$

or

$$x(x^2 - 3y^2 - 1) = 0$$

This is equivalent to two equations

$$x = 0$$
$$x^2 - 3y^2 - 1 = 0$$

The latter is a hyperbola whose asymptotes are

$$y = \pm \frac{x}{\sqrt{3}}$$

The imaginary curves are defined by

$$v = 3x^2y - y^3 - y = 0$$

This again is equivalent to two equations

$$y = 0$$
$$3x^2 - y^2 = 1$$

The latter is again a hyperbola, but this time with asymptotes

$$y = \pm\sqrt{3}x$$

The figure we have drawn (Fig. 5.5.1) looks reasonable. In the first place, at each simple zero the real and imaginary curves cross at right angles as they should according to theory. Secondly, far out, that is, going around a circle of large radius, the curves look as if they were from a triple zero; and the farther out we go, the more the local effects of the exact location of the zeros tend to fade out.

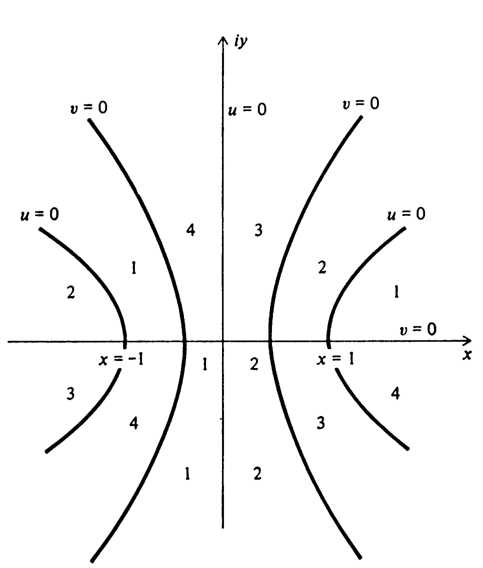


FIGURE 5.5.1

For the second example, we move the zero from $z = -1$ to $z = 0$, which makes a double zero at $z = 0$. The polynomial is

$$\begin{aligned} w &= z^2(z-1) = z^3 - z^2 \\ &= (x+iy)^3 - (x+iy)^2 \\ &= x^3 - 3xy^2 - x^2 + y^2 + i(3x^2y - y^3 - 2xy) \end{aligned}$$

The real curve is

$$u = x^3 - 3xy^2 - x^2 + y^2 = 0$$

Solving for y^2, we have

$$y^2 = \frac{x^2(x-1)}{3x-1}$$

which is easily plotted as it has a pole at $x = \frac{1}{3}$, zeros at 0 and 1, and symmetry about the x axis. As we expect, the asymptotes are parallel to

$$y = \pm \frac{x}{\sqrt{3}}$$

The imaginary curve is

$$v = 3x^2y - y^3 - 2xy = 0$$

which is

$$y(3x^2 - y^2 - 2x) = 0$$

or

$$y = 0$$

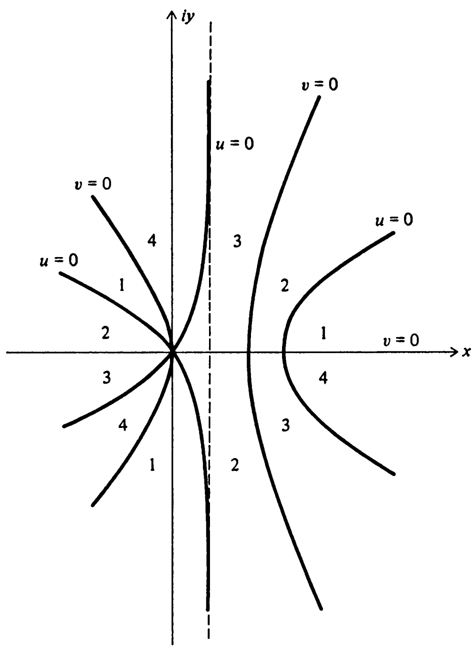


FIGURE 5.5.2

and

$$3(x - \tfrac{1}{3})^2 - y^2 = \tfrac{1}{3}$$

The last curve has asymptotes

$$y = \pm\sqrt{3}(x - \tfrac{1}{3})$$

which is what we expect.

Sketching these curves (Fig. 5.5.2), we see that far out in the complex plane they are the same as in the previous example. At the double zero there are two real curves and two imaginary curves alternating and crossing at 45° angles. The rest of the curves look as if the real and imaginary lines were being forced together by the movement of the zero from $z = -1$ to $z = 0$, and as if they tend to repel each other strongly.

Note that in the example in Sec. 5.3 the infinite sequence of zeros must be considered in judging the reasonableness of the shapes of the curves.

5.6 GENERAL RULES FOR THE $u = 0$ AND $v = 0$ CURVES

We cite without proof *the principle of the argument* which comes from complex-variable theory. This principle states that as you go around any contour, rectangular or not, in a counterclockwise direction, you will get a progression of quadrant numbers like

$$1, 1, 1, 2, 2, 2, 3, 3, 4, 4, 1, \ldots$$

with as many complete cycles 1, 2, 3, 4 as there are zeros inside (we are assuming that there are no poles in the region in which we are searching). There may at times be retrogressions in the sequence of quadrant numbers such as

$$1, 1, 1, 2, 2, 3, 3, 2, 3, 3, 4, 4, 4, 1, \ldots$$

for some suitably shaped contour, but the total number of cycles completed is exactly the number of zeros inside.

We have glossed over the instances of jumping over a quadrant number (say a 1 to a 3), and we will take that up later. We have assumed that the 0s that may occur are simply neglected, as they do not influence the total number of complete cycles.

To understand this principle of the argument, the reader can try drawing various closed contours in the previous examples. No matter how involved he draws them, he will find that he will have the correct number of zeros inside when he counts either $+1$ if he circles the zero in the counterclockwise direction or -1 if he circles the zero in the clockwise direction. Double zeros count twice, of course.

In the general analytic function, the $u = 0$ and $v = 0$ curves can be tilted at an angle to the coordinate system. They can be somewhat distorted and involved, but they must obey the three following restraints:

1 At a zero the curves must cross alternately and be spaced according to the multiplicity of the zero.

2 Far away from any zeros, the local placement of the zeros must tend to fade out and present the pattern of an isolated multiple zero having the number of all the zeros inside (with their multiplicities).

3 The number of cycles of 1, 2, 3, 4 going counterclockwise along *any* closed contour must equal the number of zeros inside that contour (when counted properly).

These three conditions so restrict the behavior of the curves we are following that many pathological situations are eliminated and the problem is thus made tractable.

PROBLEMS

5.6.1 Sketch the curves for the polynomial having zeros at

$$z = i \qquad z = i \qquad z = 0$$

5.6.2 Sketch the curves for

$$w = z^4 + 1$$

5.6.3 Sketch the curves for

$$w = z^4 + 2z^2 + 1$$

5.7 THE PLAN FOR AN IMPROVED SEARCH METHOD

One of the main faults of the crude method is that it wastes a great deal of machine time in calculating the function values (and corresponding quadrant numbers) at points which lie far from the $u = 0$ and $v = 0$ curves and hence give relatively little information.

Instead of filling in the whole area of points as we did, we propose to trace out only the $u = 0$ curves and to mark where they cross the $v = 0$ curves which gives, of course, the desired zeros. See the example at the end of Sec. 5.3 for how this might appear if we track *both* the $u = 0$ *and* the $v = 0$ curves.

The basic search pattern is to go counterclockwise around the area we are examining and look for a $u = 0$ curve, which will be indicated by a change from quadrant number 1 to 2 (or 2 to 1) or else from 3 to 4 (or 4 to 3). See Figs. 5.7.1 and 5.7.2.

When we find such a curve, we will track it until we meet a $v = 0$ curve, which is indicated by the appearance of a new quadrant number other than the two we were using to track the $u = 0$ curve.

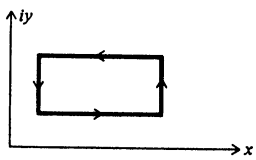


FIGURE 5.7.1

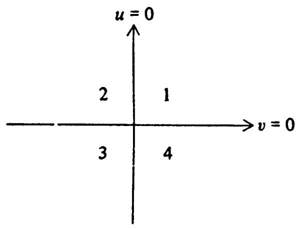


FIGURE 5.7.2

Having found the general location of a zero, we will pause to refine it, but we will need to pass over the zero finally to continue tracking our $u = 0$ curve until it goes outside the area where we are searching for zeros.

We need to know if this plan will probably locate all the zeros ("probably" depending on the step size we are using and not on the basic theory behind the plan). By the principle of the argument, the number of cycles we find is the number of zeros inside the region. Thus, the $u = 0$ and $v = 0$ curves we care about *must* cross the boundary of the region—they cannot be confined within the region—and our search along the boundary will indeed locate all the curves we are looking for (unless the step size of the search is too large) (Fig. 5.7.3).

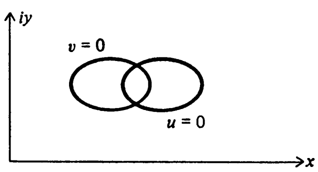


FIGURE 5.7.3 Impossible

It may happen that occasionally there is a jump in the quadrant numbers, say from 1 to 3. We can easily see that we have in one step crossed both a $u = 0$ and a $v = 0$ curve. Just where these two curves cross is, of course, not known, though probably it is near the edge (Fig. 5.7.4). If we want to find this zero, then we assume that it is a 1, 2, 3; whereas, if we wish to ignore it, we assume that it is a 1, 4, 3. We have to modify our search plan accordingly, but this is a small detail that is not worth going into at this point.

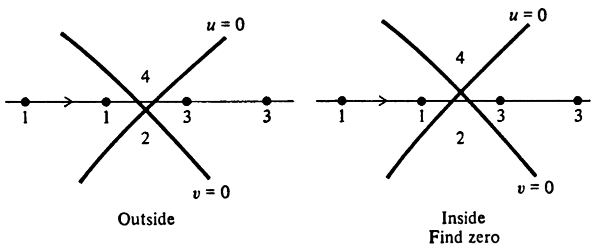


FIGURE 5.7.4

5.8 TRACKING A $u = 0$ CURVE

How shall we track a $u = 0$ curve? For convenience we start at the lower left-hand corner of the rectangular region we are examining for zeros and go counterclockwise, step by step, looking for a change in quadrant numbers that will indicate that we have crossed a $u = 0$ curve (Fig. 5.8.1). When we find such a change, we construct a square[1] inside our region, using the search interval as one side.

The $u = 0$ curve must exit from the square so that a second side of the square will have the same change in quadrant numbers. We continue in this manner,

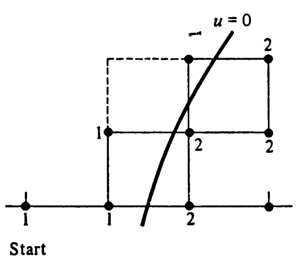


FIGURE 5.8.1

[1] Squares only if x and y are comparable in size or importance (or both); otherwise, suitably shaped rectangles to maintain relative accuracy.

each time erecting a square on the side that has the quadrant number change, until we find that a different quadrant number appears.

When this occurs, we know that we have crossed a $v = 0$ curve (Fig. 5.8.2) and that we are therefore near a complex zero of the function we are examining.

By eye it is easy to make the new square, but it is more difficulty to write the details of a program that properly chooses the two points of the next square to be examined. It is also necessary at each stage to check whether the curve we are tracking has led us out of the region we are searching, and if so, we must mark the exit (Fig. 5.8.3) so that when we come to it at a later time (while we are going around the contour), we do not track this same $u = 0$ curve again, this time in the reverse direction of course.

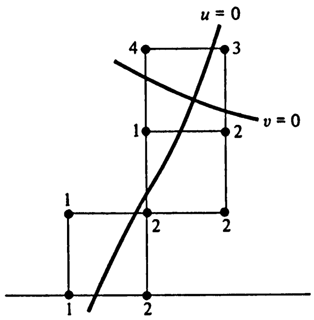


FIGURE 5.8.2

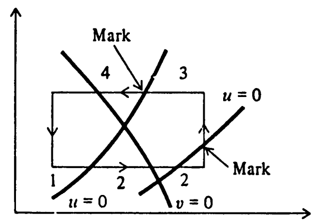


FIGURE 5.8.3

PROBLEMS

5.8.1 In the example of Sec. 5.3, apply the tracking method to the complex zero in the first quadrant.

5.8.2 Find the complex zeros as in the example in Sec. 5.3, except in the region $0 \leq x \leq 2\pi$, $2\pi \leq y \leq 4\pi$.

5.9 THE REFINEMENT PROCESS

Having located a square during our tracking of a $u = 0$ curve which has three distinct quadrant numbers, we have the clue that we are near a zero. We need, therefore, to refine our search pattern and locate the zero more accurately. The simplest way to do this is to bisect the starting side of the square that first produced the three different quadrant numbers.

If we use this smaller size, the $u = 0$ curve crosses one of the two halves, and selecting this half, we erect a square (of the new size). We may or may not find a third quadrant number. If we do not, then we continue the search with the new step size.

Usually within three steps (Fig. 5.9.1) we again have a square with three distinct quadrant numbers. We again halve and repeat the process, continuing until we have as small a square as we please (or run into either roundoff or the quantum size of the machine's number system).

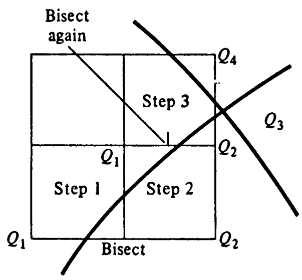


FIGURE 5.9.1

Evidently when we stop, we know that the zero is either in the square we have isolated or, at worst, in an adjacent square (again assuming that our step sizes are so small that the curves we are dealing with are reasonably smooth and well behaved).

What to do when you meet a zero value while tracking one of the curves is a small coding nuisance and is not a basic difficulty.

PROBLEMS

5.9.1 Discuss how to handle a zero value in the refinement process.

5.9.2 Discuss the use of other methods than bisection for the refinement process.

5.10 MULTIPLE ZEROS IN TRACKING

So far we have tacitly assumed that while we were tracking a $u = 0$ curve, we would come to a simple, isolated zero. But what happens when there is a double (or two very close—close for the step size we are using at the moment) or higher-order zero?

At a double zero we may find that the two quadrant numbers on the far side from the initial side of the square have the same quadrant numbers but are reversed (see Fig. 5.10.1). Thus we need, in fact, not only to check whether we have found a new quadrant number but also to check *at each step* that the same numbers are not in diagonally opposite corners, indicating two zeros or a double zero.

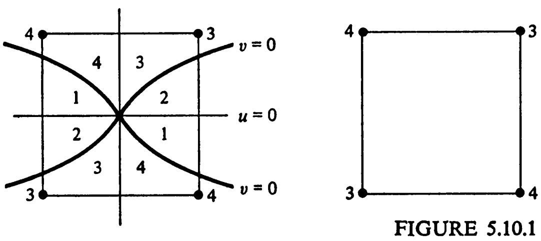


FIGURE 5.10.1

Once we sense that we are near a multiple zero, we are reasonably well off, because we can then afford relatively elaborate computing to clarify the matter. In principle it is only necessary to go around the suspected location with a sufficiently fine mesh of points to find the change in the argument and hence the

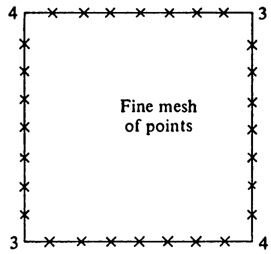


FIGURE 5.10.2

number of zeros inside (Fig. 5.10.2). Once we know this, we can proceed as the situation suggests. One way is to regard it as a new problem with a much finer search size and make an "enlargement" of the region.

Another method, which somewhat simplifies the problem of multiple zeros, is not to search one curve at a time but to go across the bottom side and find all the intervals that contain a $u = 0$ curve (Fig. 5.10.3). Then we push the calculations

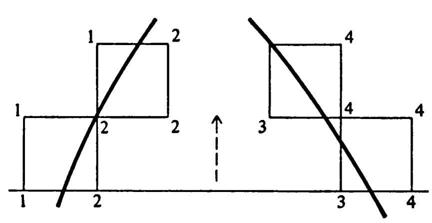


FIGURE 5.10.3

up one square, regardless of how many squares sidewise they go (and watch the sides for new $u = 0$ curves).

It is only when two or more $u = 0$ curves meet that we need to worry about a multiple zero. Thus, unless the second $u = 0$ curve is parallel to the bottom side of the rectangle, or so nearly so as to come as a total surprise, we have a very simple clue as to when to suspect a multiple zero. It may be that the two are meeting head on and are the ends of a single curve, which is not hard to detect; otherwise, we are reasonably well off in making an estimate of the number of $u = 0$ curves that are coming into the local region and hence of the multiplicity of the zero.

This method of pushing up one row at a time has the added advantage that it eliminates a lot of testing to see if the curve we are pursuing is leading us out of the region we are searching.

But let us be clear about one thing: We are vulnerable to being fooled by having chosen too large a search step, just as in the bisection method, and so the method cannot be made absolutely foolproof. What we want is a reasonably economical and safe method. All we need is a warning that there is a complicated situation at or near the location, and then we can simply apply our microscope via the enlarging process to isolate what is going on; we can repeat this enlargement process until we run into the granularity of the number system of the computing machine.

5.11 FUNCTIONS OF TWO VARIABLES

The problem of finding the real simultaneous solutions of

$$g(x, y) = 0$$
$$h(x, y) = 0$$

can be partially mapped onto the problem of finding the complex zeros of $f(x + iy) = 0$ by assigning the quadrant numbers in the obvious way:

If	Quadrant
$g > 0, h > 0$	1
$g > 0, h < 0$	2
$g < 0, h < 0$	3
$g < 0, h > 0$	4

The crude method will work, though we are not sure that the zero curves will meet at right angles at a simple zero. If we try the tracking method, we are not sure (since the principle of the argument need not apply) that we shall find all the curves as we trace the boundary. Still, the method will sometimes give useful results. Fig. 5.7.3 is now possible.

PROBLEM

5.11.1 Discuss the motion of the zeros of $w = e^{ax} - x^2 = 0$ as functions of the parameter a.

6

*ZEROS OF POLYNOMIALS

6.1 WHY STUDY THIS SPECIAL CASE?

We have just discussed the location of both real and complex zeros of a function, and it is natural to ask why we should examine the special case of polynomials,

$$w = P(z) = a_N z^N + a_{N-1} z^{N-1} + \cdots + a_0$$

This can be answered in a number of different ways.

1 The problem occurs frequently, and special methods can save a great deal of machine time and trouble.

2 For polynomials we can apply the invariance principle very effectively because we have a large class of transformations that leave the problem essentially unchanged.

3 A polynomial of degree N has exactly N zeros, and we therefore know when we have found all the zeros.

4 The factor theorem shows that in the polynomial case, zeros and factors are equivalent, and we can use the divisibility as a tool in finding the zeros.

5 When we find a zero, we can "deflate" the polynomial by dividing out this factor and thus obtain a simpler problem to solve.
6 In applications it is often of crucial importance to identify multiple zeros as multiple zeros and not as close ones.
7 In applications it is often true that the zeros are known to be real and/or pure imaginary and those that fall off the two axes were moved by roundoff. Therefore, the polynomial routine can "nudge" such zeros back onto their respective axes *provided* the difference is attributable to roundoff effects.
8 In applications it appears that often the user is more interested in the zeros as a self-consistent set rather than as a set of individually accurate, but unrelated, zeros.
9 From the factors we find, we can reconstruct the polynomial (approximately) and compare these coefficients with the original to form some measure of the loss of information around the whole loop. Since the reconstruction is fairly direct, presumably most of the loss occurs while finding the factors. (Compare Sec. 3.2 where we discussed quadratics.)

The point about multiple zeros needs elaboration. If a double zero is reported as a pair of close zeros, then when they are used it is highly likely that there will be great trouble with roundoff. We have adopted the attitude that we prefer to avoid roundoff rather than bound it by an elaborate rule, and this means that we must make special efforts to identify multiple zeros so that later the roundoff will not cause trouble.

As an example of the effect of a double zero being reported as a pair of close ones, consider the problem of solving the simple linear differential equation

$$y'' + 2y' + y = 0 \qquad y(0) = 1 \qquad \text{and} \qquad y'(0) = 0$$

The characteristic equation is

$$\begin{aligned} m^2 + 2m + 1 &= 0 \\ (m+1)^2 &= 0 \\ m &= -1, -1 \end{aligned}$$

The solution is, therefore,

$$\begin{aligned} y &= e^{-x}(C_1 + C_2 x) \\ &= e^{-x}(1 + x) \end{aligned}$$

But if due to roundoff the roots were given as

$$\begin{aligned} m_1 &= -1 + \varepsilon \\ m_2 &= -1 - \varepsilon \qquad \varepsilon \text{ small} \end{aligned}$$

then the general solution would be

$$y = C_1 e^{-(1-\varepsilon)x} + C_2 e^{-(1+\varepsilon)x}$$

Applying the initial conditions we get

$$C_1 + C_2 = 1$$
$$-(1-\varepsilon)C_1 - (1+\varepsilon)C_2 = 0$$

and

$$y = \frac{1}{2}\left[\left(1+\frac{1}{\varepsilon}\right)e^{-(1-\varepsilon)x} - \left(\frac{1}{\varepsilon}-1\right)e^{-(1+\varepsilon)x}\right]$$

Note the roundoff trouble, especially near $x = 0$.

As a second example of this important effect consider the integration of rational functions. If the zeros of the denominator were multiple but were reported as close, then instead of

$$\int_0^\infty \frac{dx}{(a^2+x^2)^2} = \int_0^{\pi/2} \frac{a\sec^2 t}{a^4 \sec^4 t}\,dt = \frac{1}{a^3}\int_0^{\pi/2} \cos^2 t\,dt = \frac{\pi}{4a^3}$$

we would have the integral (with b close to a)

$$\begin{aligned}\int_0^\infty \frac{dx}{(x^2+a^2)(x^2+b^2)} &= \frac{1}{b^2-a^2}\int_0^\infty \left(\frac{1}{x^2+a^2} - \frac{1}{x^2+b^2}\right)dx \\ &= \frac{1}{b^2-a^2}\left(\frac{1}{a}\arctan\frac{x}{a} - \frac{1}{b}\arctan\frac{x}{b}\right)\Bigg|_0^\infty \\ &= \frac{\pi}{2}\frac{1}{b^2-a^2}\left(\frac{1}{a}-\frac{1}{b}\right)\end{aligned}$$

This expression can be reworked, using the methods of Chap. 3, to avoid the obvious heavy cancellation. In this simple example the trouble is obvious and typical of the general case of integration of rational functions when a multiple zero occurs in the denominator.

PROBLEMS

6.1.1 Devise another example to demonstrate the effect of close zeros on subsequent computations.

6.1.2 Discuss the case

$$\int_0^a \frac{dx}{(a+x)^3} = \frac{3}{8a^2} \qquad a > 0$$

Hint: use $-a$, $-(a-\varepsilon)$, $-(a+\varepsilon)$ as zeros.

6.2 INVARIANCE PRINCIPLE

Polynomials provide a good example of the use of the invariance principle because we have three different transformations that leave the polynomial essentially unchanged:

1 The transformation of $P(z)$ into $cP(z)$
2 The transformation of $P(z)$ into $P(cz)$
3 The transformation of $P(z)$ into $z^N P\left(\frac{1}{z}\right)$

The special case under (2), where $c = -1$, reverses the direction of the x axis and can be used, if we wish, to confine the search to finding positive zeros only.

The reciprocal transformation (3) has the effect of reversing the coefficients of the polynomial. This is mirrored in the synthetic division process. We have written the polynomial (a linear combination of successive powers of z) as if $|z| > 1$ and have put the highest power of z first, much as we would when writing numbers. But if we think of z as being less than 1 in size, then we would naturally write it as

$$a_0 + a_1 z + a_2 z^2 + \cdots + a_N z^N$$

and in the division process divide by (for a quadratic factor)

$$1 - p'z - q'z^2$$

The remainder would be

$$r'_{N-1} z^{N-1} + r'_N z^N$$

If in the usual process the divisor were a factor of the polynomial, then the remainders r_1 and r_0 would both be zero; in the second case it would be r'_{N-1} and r'_N that would be zero. A little thought shows that these two are equivalent and that the two quotients *must be the same*. Thus we have two alternate ways of using the third transformation: either we can reverse the coefficients of the polynomial (which carries the region $1 \leq |z| < \infty$ into $1 \geq |z| > 0$) or else equivalently we can do our synthetic division in the reverse way, from constant to higher powers rather than the more conventional order of higher to constant powers.

We shall check regularly that the methods we propose to use obey the invariance principle with respect to these three transformations.

PROBLEMS

6.2.1 The product of a number of factors $(x - c_1)(x - c_2) \cdots (x - c_N)$ leads to a polynomial whose coefficients are the elementary symmetric functions

$$E_1 = \sum c_i,\ E_2 = \sum c_i c_j,\ \ldots,\ E_N = c_1 c_2 \cdots c_N$$

Thus for

$$x^N + a_{N-1}x^{N-1} + \cdots + a_0$$

with zeros $c_1, c_2, \ldots, c_N$,

$$a_{N-k} = (-1)^k E_k$$

Discuss the roundoff errors in constructing this polynomial.

6.2.2 Devise a measure of closeness of two polynomials using only the coefficients. Defend your choice.

6.3 THE PLAN

One of the consequences of the complex zeros of a real function occurring in pairs is that by the fundamental theorem of algebra a polynomial with real coefficients can be written as a product of real linear and real quadratic factors. This approach has the advantage, as shown in Sec. 3.9, of working only with real numbers and of avoiding special programming for complex numbers (also saving machine time). The approach to the problem of finding the factors has the further advantages of ease of thought and of finding what is most often needed in the next step, namely, the factors rather than the actual zeros.

Therefore, after examining the given coefficients to see what kind of polynomial we have and making the obvious possible simplifications, we will first find all the real linear factors, watching carefully for multiple factors, and deflate the polynomial each time we find one. Then we will search for the real quadratic factors, again watching for multiple factors and deflating each time we find a factor. Further, each time we find a quadratic factor, we will check to see if the zeros of the factor might have been on one of the axes and had been moved off due to roundoff.

The methods we use will be guided by the requirements of the invariance principle.

6.4 PREPROCESSING THE POLYNOMIAL

Often the polynomial which we are given is said to be of degree N, but is not. For example, the leading coefficient a_N may be zero (perhaps for a particular value of a parameter in the equation), which means that the polynomial has a zero at infinity. Further it may be that the last coefficient a_0 is zero, which means that there is a zero at the origin. In both cases these zeros should be removed before starting. To do this we can test for

$$a_0 a_N = 0$$

and if the relation holds, we must find out which of the factors is zero and remove the corresponding zero from the polynomial. This test applied recursively will remove all these trivial zeros.

Next we need to find out if it is a polynomial in some power of z, say z^2 or z^3, and if it is, we should make the proper substitution to reduce the effective degree of the polynomial. For example, the polynomial

$$x^{12} - 6x^8 + 4x^4 + 1 \equiv (x^4)^3 - 6(x^4)^2 + 4(x^4) + 1$$

is actually a cubic in x^4 and should be treated as a cubic and not as a polynomial of twelfth degree.

The factor k that we want is the common factor of all the exponents of terms with nonzero coefficients. To find this factor we start with the constant term $a_0 \neq 0$ and look for the next higher term with a nonzero coefficient. If it is a_1, we are done—$k = 1$; but if it is some higher power, we find the factors of the exponent which are also factors of the degree N. Each factor that divides N is a potential candidate for further search to see if all terms in the polynomial have exponents that are multiples of it. In this way we can find the highest common factor of the exponents those terms present in the polynomial.

Suppose we have found such a number $k > 1$. Making the obvious substitution,

$$z^k = z'$$

we have a polynomial in z' of much lower degree to solve, one having no common factors of the exponents of the terms present.

This transformation must, of course, be undone when we finally produce the answers. For the real linear factors

$$z' = a = z^k$$

Write a in the polar form

$$a = \rho e^{i\theta} \qquad \text{where} \qquad \rho = |a| \qquad \theta = \begin{cases} 0 & \text{for } a > 0 \\ \pi & \text{for } a < 0 \end{cases}$$

$$z^k = \rho e^{i(\theta + 2\pi m)} \qquad m = 0, 1, \ldots, k - 1$$

Taking the kth root, we have the k zeros

$$z_m = \rho^{1/k} e^{i[(\theta + 2\pi m)/k]} = \rho^{1/k}\left(\cos\frac{\theta + 2\pi m}{k} + i \sin\frac{\theta + 2\pi m}{k}\right)$$

$$m = 0, 1, \ldots, k - 1$$

For a quadratic factor

$$(z')^2 - pz' - q = z^{2k} - pz^k - q \qquad q < 0$$

write it in the form

$$(z^k - \rho e^{i\theta})(z^k - \rho e^{-i\theta}) = z^{2k} - (2\rho \cos \theta)z^k + \rho^2$$

so that

$$\begin{aligned} \rho^2 &= -q & & \rho = \sqrt{-q} \\ & & \text{or} & \\ 2\rho \cos \theta &= p & & \theta = \arccos \frac{p}{2\rho} \end{aligned}$$

We therefore have the k real quadratic factors

$$(z - \rho^{1/k} e^{i[(\theta + 2\pi m)/k]})(z - \rho^{1/k} e^{-i[(\theta + 2m\pi)/k]})$$

$$\equiv z^2 - 2\rho^{1/k} \cos\left(\frac{\theta + 2m\pi}{k}\right) z + \rho^{2/k} \qquad m = 0, 1, \ldots, k-1$$

or

$$\begin{cases} p_m = 2\rho^{1/k} \cos \dfrac{\theta + 2m\pi}{k} \\ q_m = -\rho^{2/k} \end{cases} \qquad m = 0, 1, \ldots, k-1$$

6.5 THE REAL ZEROS

We first remove the real zeros. For this purpose we can use either the bisection or the modified false-position method. In both cases we make an initial search for intervals with a sign change. When we find a sign change, we can refine the interval as much as we wish, and then we deflate the polynomial.

Now the product of all the zeros is

$$(-1)^N \frac{a_0}{a_N}$$

so that the geometric mean of all the zeros

$$r = \sqrt[N]{\left|\frac{a_0}{a_N}\right|}$$

provides a natural unit of length. The invariance principle tells us that since the reciprocal transformation and the reversal of the negative and positive axes leave the polynomial unchanged, we must treat the four ranges

$$0 \le z \le r$$
$$0 \ge z \ge -r$$
$$0 \le \frac{1}{z} \le \frac{1}{r} \qquad \text{where } z \text{ is a real number}$$
$$0 \ge \frac{1}{z} \le -\frac{1}{r}$$

in essentially the same way. As a matter of engineering judgement we will search each range, using $2N$ equally sized steps, and take one step in each range before going to the next step. We start searching at zero because we prefer the risk of an underflow to an overflow and because we instinctively feel that the roundoff propagation will be less. Thus, except for trivial details the process of searching is invariant under the three transformations,[1] since the refinement method was invariant under the first two and the third is covered by breaking the problem into four ranges.

We need, of course, to watch for multiple zeros. What do we mean by a zero of multiplicity k? Since a polynomial of degree N can be factored into N factors (not necessarily distinct), it is conventional (because the remainder theorem of Sec. 3.7 connects the zeros and factors of a polynomial) to identify *each* factor with a zero and to say that a polynomial of degree N has exactly N zeros. This is apparently how the idea of multiple zeros arose.

The Taylor-series expansion of a function gives a second approach. The number of successive coefficients (of the now possibly infinite sequence of coefficients) that vanish at a point is the multiplicity of the zero. In this fashion the idea of multiplicity, defined originally for polynomials, is extended to analytic functions. At this point we do not need the further extension to functions such as the very common

$$y = \sqrt{a^2 - x^2}$$

which could be said to have a real zero of order $\frac{1}{2}$ at $x = \pm a$.

The repeated synthetic division process in Sec. 3.7 identified $k!$ times the successive remainders with the kth derivative. While we are searching for a zero by either the bisection or modified false position, we will evaluate the successive derivatives together with the estimates of the roundoff noise to find which is the highest-order derivative that can be regarded as vanishing. Using the next remainder in the sequence (which is therefore not zero), we make the preceding

[1] It is easy to devise an algorithm which uniquely tells in which order the four ranges are to be searched, but it does not appear worth using in practice.

one as small as we can (we could use Newton's method here if we pleased). For a possible double zero, we are finding the place where the derivative is zero, for a triple zero, the local inflection point, etc., which seems to be quite a reasonable way to place the multiple zero. We need, of course, to check that the original zero is still within roundoff, and if it is not, we then have guessed at too high a multiplicity and must go back and try again.

In our search process we watch not only for sign changes in the function values, but also to be safe we watch for sign changes in the first derivative. Together these isolate both the odd- and even-order zeros (probably).

$$
\begin{array}{l|lll}
x & x\ x\ x \ldots x\ x\ x & & \\
 & \quad x\ x \ldots x\ x\ x & & \\ \hline
x & x\ x\ x \ldots x\ x & x \sim 0 & \\
 & \quad x\ x \ldots x\ x & & \\ \hline
x & x\ x\ x \ldots x & x \sim 0 & \\
 & \quad x\ x \ldots x & & \\ \hline
 & x\ x\ x \ldots & x \neq 0 &
\end{array}
$$

Use this

to make this zero.

PROBLEMS

6.5.1 Prove that even in the presence of roundoff when we end the search process, we will have a polynomial of even degree.

6.5.2 Devise a reasonable algorithm to carry out the remark in footnote 1 in Sec. 6.5.

6.5.3 Show that

$$\frac{1}{\rho(\cos\theta + i\sin\theta)} = \frac{\cos\theta - i\sin\theta}{\rho}$$

or

$$\frac{1}{x+iy} = \frac{x-iy}{x^2+y^2}$$

enables us to undo the reciprocal transformation.

6.6 PLAN FOR FINDING COMPLEX ZEROS

Having (probably) removed all the real zeros, we will have left a polynomial of even degree

$$w(z) = P_{2M}(z) = a_{2M}z^{2M} + a_{2M-1}z^{2M-1} + \cdots + a_0$$

Searching for real quadratic factors instead of complex zeros means that we are not searching the complex x, iy plane but the pq plane of the coefficients of the quadratic factor (see Fig. 6.6.1). These two planes are connected by

$$Q(z) = z^2 - pz - q = 0$$

This means that for both a point in the upper half of the complex plane and its conjugate point there is one point in the pq plane. The line of the x-axis goes into the curve of multiple zeros, namely when the radical is zero

$$\sqrt{p^2 + 4q} = 0$$

or more conveniently,

$$p^2 \equiv -4q$$

Inside this parabola the zeros of the quadratic factor are complex, along the curve they are multiple, and above the curve they are a pair of real distinct zeros (which

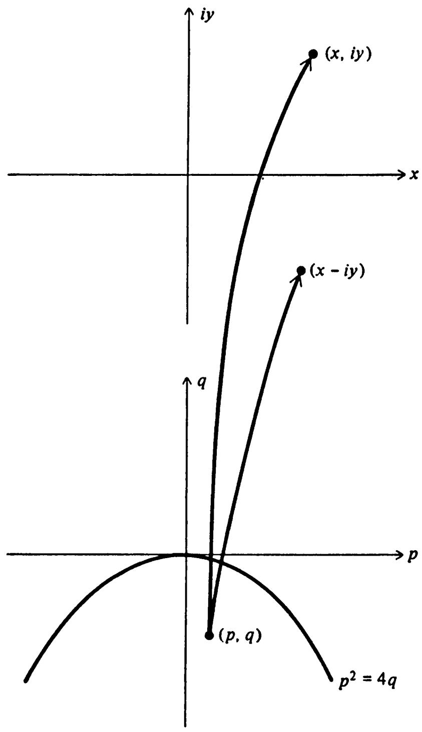


FIGURE 6.6.1

we have removed from the polynomial). Since we have deliberately searched for multiple zeros and have removed all the real zeros, we are definitely *inside* the parabola and not "near" it.

We next examine the basic Bairstow method that we will use in finding the real quadratic factors.

6.7 BAIRSTOW'S METHOD

When we divide a polynomial of even degree $P_{2M}(z)$ by a quadratic factor $Q(z) = z^2 - pz - q$, we get a quotient and a remainder

$$r_1 z + r_0$$

As in Sec. 3.7, repeated divisions of the quotient by $Q(z)$ produce an expansion in powers of $Q(z)$, namely,

$$\begin{aligned} P_{2M}(z) &= a_{2M}\, Q^M(z) \\ &\quad + (r_{2M-1}z + r_{2M-2})Q^{M-1}(z) + \cdots + (r_3 z + r_2)Q(z) + (r_1 z + r_0) \end{aligned}$$

We know that $Q(z)$ is a factor if and only if $r_1 = r_0 = 0$.

Since the remainders

$$\begin{aligned} r_1 &= r_1(p, q) \\ r_0 &= r_0(p, q) \end{aligned}$$

are analytic functions in p and q, we may expand them in a Taylor series about the current point (p, q). The value at any nearby point (p^*, q^*) in the pq plane is given by

$$\begin{aligned} r_1(p^*, q^*) &= r_1(p, q) + \frac{\partial r_1}{\partial p}\,\Delta p + \frac{\partial r_1}{\partial q}\,\Delta q + \cdots \\ r_0(p^*, q^*) &= r_0(p, q) + \frac{\partial r_0}{\partial p}\,\Delta p + \frac{\partial r_0}{\partial q}\,\Delta q + \cdots \end{aligned}$$

where

$$\begin{aligned} \Delta p &= p^* - p \\ \Delta q &= q^* - q \end{aligned}$$

As a first approximation we drop all the terms beyond the linear ones. Next we pick (p^*, q^*) so that

$$\begin{aligned} r_1(p^*, q^*) &= 0 \\ r_0(p^*, q^*) &= 0 \end{aligned}$$

will improve our guesses at p and q. Note that this is essentially Newton's method in two variables.

To get the required partial derivatives, we observe that $P_{2M}(z)$ does not depend on p and q, and we differentiate

$$\frac{\partial P_{2M}}{\partial p} \equiv \{\cdots\}Q(z) + (r_3 z + r_2)\frac{\partial Q}{\partial p} + \frac{\partial r_1}{\partial p} z + \frac{\partial r_0}{\partial p} = 0$$

$$\frac{\partial P_{2M}}{\partial q} \equiv \{\cdots\}Q(z) + (r_3 z + r_2)\frac{\partial Q}{\partial q} + \frac{\partial r_1}{\partial q} z + \frac{\partial r_0}{\partial q} = 0$$

These equations are *identities* in z, hence are true for all values of z and in particular for those which make $Q(z) = 0$. Using one of these values, we have

$$(r_3 z + r_2)(-z) + \frac{\partial r_1}{\partial p} z + \frac{\partial r_0}{\partial p} = 0$$

$$(r_3 z + r_2)(-1) + \frac{\partial r_1}{\partial q} z + \frac{\partial r_0}{\partial q} = 0$$

But since $z^2 = pz + q$ and we are operating formally, we regard z as a complex number and set the linear terms and (separately) the constant terms equal to zero.[1] Thus we get the four equations

$$r_3 p + r_2 = \frac{\partial r_1}{\partial p}$$

$$r_3 q = \frac{\partial r_0}{\partial p}$$

$$r_3 = \frac{\partial r_1}{\partial q}$$

$$r_2 = \frac{\partial r_0}{\partial q}$$

Using these, the Newton approximations (the truncated Taylor-series expansions) become the usual Bairstow equations

$$(r_3 p + r_2)\,\Delta p + r_3\,\Delta q = -r_1$$

$$(r_3 q)\,\Delta p + r_2\,\Delta q = -r_0$$

The determinant of these equations is

$$D = r_2^2 + r_2 r_3 p - r_3^2 q$$

$$= r_3^2\left[\left(\frac{r_2}{r_3}\right)^2 + p\frac{r_2}{r_3} - q\right]$$

[1] We are using the fact that if $az + b = 0$, with a and b real and z complex, then $a = b = 0$.